

The Vibrational Structure of Davydov Solitons

This content has been downloaded from IOPscience. Please scroll down to see the full text.

1982 Phys. Scr. 25 651

(<http://iopscience.iop.org/1402-4896/25/5/015>)

View [the table of contents for this issue](#), or go to the [journal homepage](#) for more

Download details:

IP Address: 130.63.180.147

This content was downloaded on 26/05/2014 at 05:40

Please note that [terms and conditions apply](#).

The Vibrational Structure of Davydov Solitons

Alwyn C. Scott

Center for Nonlinear Studies, Los Alamos National Laboratory, Los Alamos, New Mexico 87545, U.S.A.

Received December 29, 1981; accepted February 4, 1982

Abstract

Internal oscillations of an alpha-helix soliton are studied numerically on a modified version of Davydov's equations and analytically on an approximating system of coupled nonlinear Schrödinger equations. The numerical results are used to interpret lines in recently measured laser-Raman spectra of metabolically active *E. coli*. Agreement between measured and calculated lines appears to be within the uncertainties of computation and measurement.

1. Introduction

The idea that bond energy could be transported along one-dimensional molecular chains (such as alpha-helix protein molecules) by solitary waves (or "solitons") was first proposed by Davydov and Kislukha in 1973 [1]. Since that time Davydov and his co-workers have published a number of papers that discuss various soliton parameters including lifetime, damping, energy, and wave structure [1]. The physical notion of Davydov's alpha-helix soliton can be understood rather easily with reference to Fig. 1. The energy to be transported is stored as "stretching" quanta of the amide-I bond (i.e., $C=O$) which has an absorption peak at 1653 cm^{-1} corresponding to a quantum unit of 0.205 eV . Naïvely one would expect such energy to disperse rapidly ($\sim 10^{-12}\text{ s}$) via dipole-dipole interactions throughout all the amide-I bonds. But if certain nonlinearities of the system are taken into consideration, it is seen that localized bond energy generates longitudinal sound which in turn acts as a potential well to trap the bond energy and prevent its dispersion. Thus the Davydov soliton can be viewed as a local region of conformational change which is mobile.

The most important nonlinearity in the problem is a parameter (χ_1) that gives the change in energy of an amide-I bond (newton-meters) per unit extension of its associated hydrogen bond (meters). In 1979 a theoretical study of an analytically tractable approximation to Davydov's equations (the nonlinear Schrödinger equation) showed that there is a threshold level for χ_1 below which solitons could not be expected to form, and this conclusion was supported by numerical integration of Davydov's equations [3]. Estimates by both Davydov and myself indicated for a real alpha-helix protein that χ_1 was about the right order of magnitude to support soliton formation.

Recently Kuprievich and Kudritskaja [4] have obtained χ_1 from SCF calculations on formamide dimer as 0.34×10^{-10} newtons. Using a form of Davydov's equations that have been extended to improve accuracy, I have recently found exactly the same value of χ_1 as the threshold for soliton formation [5]. Thus there is now substantial numerical evidence that Davydov solitons can form on real alpha-helix protein.

During the course of this study [5] some cursory

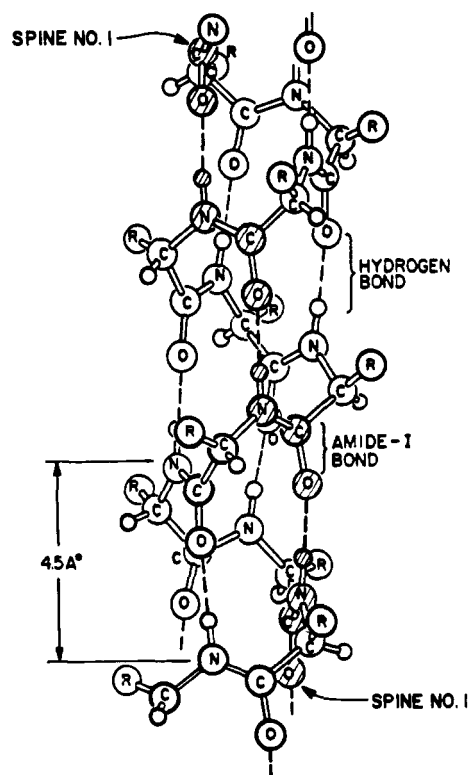


Fig. 1. Alpha-helix (redrawn from [2] with permission of the author). The atoms of a single "spine" are cross-hatched.

observations were made of an "interspine oscillation" in bond energy between the three spines (etc., $\cdots\text{H}-\text{N}-\text{C}=\text{O}\cdots\text{H}-\text{N}-\text{C}=\text{O}\cdots$, etc.) that run longitudinally along the helix. (One of these three spines is cross-hatched in Fig. 1). After the manuscript for [5] was completed, I became aware of Webb's recent review of laser-Raman spectra measured on metabolically active cells [6]. Of particular interest was a spectrum (see Fig. 2) taken from *E. coli* and consisting of nine definite lines which could be assigned rather closely to simple sums and differences of two predetermined frequencies: (i) The frequency of interaction of the soliton with unit cells of the alpha-helix, and (ii) The frequency of the above mentioned interspine oscillation [7]. Thus laser-Raman spectroscopy (LRS) may provide direct experimental confirmation that Davydov solitons play a functional role in metabolic processes. If so, LRS could become even more valuable as a probe into the secret details of life processes.

The aim of this paper is to provide the basis for a critical evaluation of the above hypothesis (that Fig. 2 displays the vibrational structure of an alpha-helix soliton). The following section lists the numerical and analytical models to be studied beginning with that developed in [5] and proceeding toward simpler and more tractable approximations. Section 3 discusses the linear problem of interspine interaction below the level for soliton formation. This is important not only because the linear

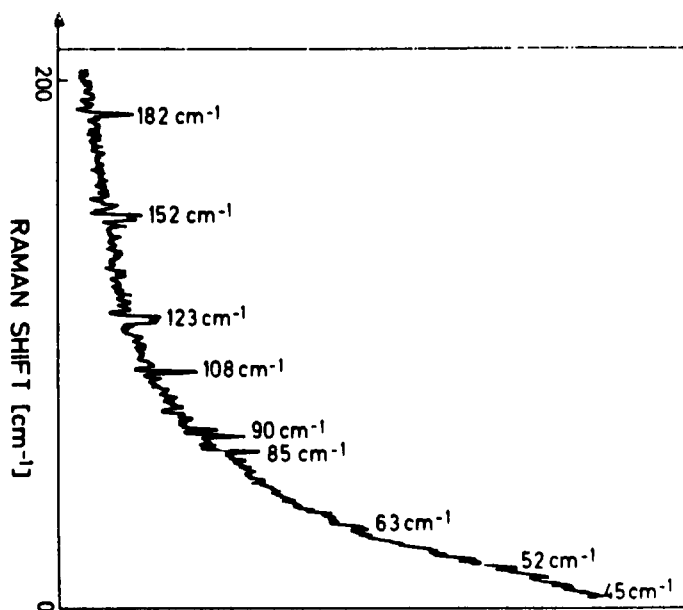


Fig. 2. Laser-Raman spectrum from metabolically active *E. coli* (from [6] with permission of the author).

problem suggests ways to analyze the nonlinear problem above threshold, but also because it is necessary to understand the dynamic differences above and below threshold in order to consider whether Fig. 2 represents the spectrum of a soliton or mere linear vibrations of the protein. In Section 4 the simplest representation of the nonlinear problem is considered: Three coupled nonlinear Schrödinger (NLS) equations. At this level of approximation, a family of exact soliton solutions is obtained which display the salient features of the solitary wave observed on the original numerical model. A detailed description of this numerical solitary wave is presented in Section V and discussed in the context of the analytical solution developed in the previous section. The differences between nonlinear system dynamics above threshold and linear dynamics below threshold are also developed. A critical discussion of the differences between the spectrum of Fig. 2 and my numerical prediction of it is presented in Section VI. In particular, it is shown that the best possible numerical results differ only slightly from those obtained from the numerical model without any adjustment of the physical parameters. The residual differences between predicted and measured spectra must be attributed to errors in measurement. Section 7 provides a concluding discussion. The analytical results of Section 4 are readily extended from three to an arbitrary number of interacting NLS equations; this extension is noted in an appendix.

2. Alpha-helix models

In this section the several numerical and analytical models to be considered are listed in order of decreasing complexity.

2.1. Modified Davydov (MD) equations

The following equations were developed in [5] to describe the coupled propagation of bond energy probability density, $A_{n\alpha}$, and longitudinal displacement of an amide-I bond, $B_{n\alpha}$.

$$i \frac{dA_{n\alpha}}{d\tau} + J(A_{n+1,\alpha} - 2A_{n\alpha} + A_{n-1,\alpha}) - k_1(B_{n+1,\alpha}$$

Table I

Term	Energy (computer units)
J	0.145
L	0.231
N	0.073
P	0.034
Q	0.019
R	0.012
S	0.0089
T	0.0073
U	0.0037
V	0.0030
X	0.0022
Z	0.0017

$$\begin{aligned} -B_{n-1,\alpha})A_{n\alpha} = & LF_L + \{k_2[B_{n-1,\alpha}A_{n-1,\alpha} \\ & - B_{n+1,\alpha}A_{n+1,\alpha} + B_{n\alpha}(A_{n+1,\alpha} - A_{n-1,\alpha})] - 2JA_{n\alpha} \\ & + NF_N + PF_P + QF_Q + RF_R + SF_S + TF_T + UF_U \\ & + VF_V + XF_X + ZF_Z\} \end{aligned} \quad (\text{MD1})$$

$$\begin{aligned} \frac{d^2 B_{n\alpha}}{d\tau^2} - (B_{n+1,\alpha} - 2B_{n\alpha} + B_{n-1,\alpha}) = & k_3(|A_{n+1,\alpha}|^2 \\ & - |A_{n-1,\alpha}|^2) + \{k_4[A_{n\alpha}^*(A_{n+1,\alpha} - A_{n-1,\alpha}) \\ & + A_{n\alpha}(A_{n+1,\alpha}^* - A_{n-1,\alpha}^*)]\} \end{aligned} \quad (\text{MD2})$$

In these equations the $\{J, L, \dots, Z\}$ are dipole-dipole interaction energies between pairs of amide-I bonds. They have the normalized values given in Table I. The subscript n counts unit cells along the helix while α specifies one of the three spines (see Fig. 1). The symbol F_L is shorthand for the dipole-dipole interactions between laterally adjacent amide-I bonds; thus:

$$\begin{aligned} i\dot{A}_{n1} &= L(A_{n2} + A_{n-1,3}) + \text{etc.} \\ i\dot{A}_{n2} &= L(A_{n3} + A_{n1}) + \text{etc.} \\ i\dot{A}_{n3} &= L(A_{n2} + A_{n+1,1}) + \text{etc.} \end{aligned} \quad (1a, b, c)$$

Similar definitions of the symbols $F_N, \dots, F_Z$ are given in [5].

The constants $k_1, \dots, k_4$ represent nonlinear coupling between bond energy dispersion and longitudinal sound. They are

$$\begin{aligned} k_1 &= \frac{10^{-11}}{\hbar} \sqrt{\frac{M}{w}} \chi_1 \\ k_2 &= \frac{10^{-11}}{\hbar} \sqrt{\frac{M}{w}} \chi_2 \\ k_3 &= \frac{10^{11}}{w} \chi_1 \\ k_4 &= \frac{10^{11}}{w} \chi_2 \end{aligned} \quad (2a, b, c, d)$$

where M is one third of the mass of a unit cell, w is the linear force constant of a hydrogen bond, χ_1 (as previously noted) is the change in amide-I bond energy per unit extension of an adjacent hydrogen bond, and χ_2 is the change in the longitudinal (nearest neighbor) dipole-dipole coupling term, J , per unit of differential displacement.

The time scale, τ , measures laboratory time in units of

$$\sqrt{\frac{M}{w}} = 0.99 \times 10^{-13} \text{ s} \quad (3)$$

and $B_{n\alpha}$ measures longitudinal displacement in units of 10^{-11} m .

It is important to emphasize (see [5] for details) that there are no adjustable parameters in the MD system.

2.2. Coupled Zakharov (CZ) equations

The MD system can be simplified by noting [5] that $k_2 \ll k_1$ and $k_4 \ll k_3$ and also from Table I that J and L are substantially larger than the other dipole-dipole interaction terms. Furthermore the term “ $-2JA_{n\alpha}$ ” in MD1 can be removed by scaling. Thus the $\{ \}$ terms can be neglected in both MD1 and MD2. Also the discrete variable, n , can be approximated as a continuous variable, ξ , by writing

$$B_{n+1,\alpha} - B_{n-1,\alpha} \doteq 2 \frac{\partial \tilde{B}_\alpha}{\partial \xi}$$

$$|A_{n+1,\alpha}|^2 - |A_{n-1,\alpha}|^2 \doteq 2 \frac{\partial |\tilde{A}_\alpha|^2}{\partial \xi}$$

Finally it is convenient to define

$$\tilde{\rho}_\alpha \equiv \frac{\partial \tilde{B}_\alpha}{\partial \xi} \quad (4)$$

whereupon the MD system reduces to

$$i \frac{\partial \tilde{A}_\alpha}{\partial \tau} + J \frac{\partial^2 \tilde{A}_\alpha}{\partial \xi^2} - 2k_1 \tilde{\rho}_\alpha \tilde{A}_\alpha \doteq L(\tilde{A}_{\alpha+1} + \tilde{A}_{\alpha-1}) \quad (\text{CZ1})$$

$$\frac{\partial^2 \tilde{\rho}_\alpha}{\partial \tau^2} - \frac{\partial^2 \tilde{\rho}_\alpha}{\partial \xi^2} \doteq 2k_3 \frac{\partial^2 |\tilde{A}_\alpha|^2}{\partial \xi^2} \quad (\text{CZ2})$$

where $\alpha = 1, 2, 3$. Writing

$$\tilde{A}_\alpha = A_\alpha / 4k_1 k_3 J$$

$$\tilde{\rho}_\alpha = \rho_\alpha / 2k_1 J$$

$$\xi = Jx$$

$$\tau = Jt$$

puts CZ into the normalized form*

$$i A_{\alpha,t} + A_{\alpha,xx} - \rho_\alpha A_\alpha \doteq \epsilon (A_{\alpha+1} + A_{\alpha-1}) \quad (\text{CZ1}')$$

$$\rho_{\alpha,tt} - \rho_{\alpha,xx} \doteq (|A_\alpha|^2)_{xx} \quad (\text{CZ2}')$$

where

$$\epsilon = JL \quad (5)$$

2.3. Coupled nonlinear Schrödinger (CNLS) equations

For the solitary wave solutions that arise in this study, the velocities observed are small compared with unity (longitudinal sound speed). Thus the first term in (CZ2') can be neglected, leading to the coupled system of nonlinear Schrödinger equations.

$$i A_{1,t} + A_{1,xx} + |A_1|^2 A_1 \doteq \epsilon (A_2 + A_3)$$

$$i A_{2,t} + A_{2,xx} + |A_2|^2 A_2 \doteq \epsilon (A_1 + A_3)$$

$$i A_{3,t} + A_{3,xx} + |A_3|^2 A_3 \doteq \epsilon (A_1 + A_2) \quad (\text{CNLS})$$

2.4. Coupled nonlinear Schrödinger equations with no bending (CNLSNB)

A slightly different form of coupled nonlinear Schrödinger equations can be obtained from CZ' under the assumption that the dynamics introduces no bending of the alpha-helix. This assumption is equivalent to requiring

$$\rho_1 = \rho_2 = \rho_3 = \frac{1}{3}(|A_1|^2 + |A_2|^2 + |A_3|^2) \quad (6)$$

whereupon the three coupled nonlinear Schrödinger equations can be written in vector form as

$$i \bar{A}_t + \bar{A}_{xx} + \frac{1}{3}(\bar{A}^* \cdot \bar{A}) \bar{A} = Z \bar{A} \quad (\text{CNLSNB})$$

where $\bar{A} = \text{col}(A_1, A_2, A_3)$ and

$$Z = \begin{bmatrix} 0 & \epsilon & \epsilon \\ \epsilon & 0 & \epsilon \\ \epsilon & \epsilon & 0 \end{bmatrix}. \quad (7)$$

This “no bending” assumption is not unreasonable physically since the valence structure of the helix might be expected to inhibit differences in the ρ_α 's, but CNLSNB is primarily interesting because, as we shall see in Section 4, it is completely integrable.

2.5. Coupled linear Schrödinger (CLS) equations

As the amplitudes of $\bar{A}$ approach zero, both CNLS and CNLSNB become the coupled linear Schrödinger equations which can be written in vector form as

$$i \bar{A}_t + \bar{A}_{xx} = Z \bar{A} \quad (\text{CLS})$$

3. Linear analysis below threshold for soliton formation

A general solution of CLS can be written as a generalized sum of elementary solutions of the form $\bar{A} \exp(i\omega t + ikx)$ whereupon CLS requires

$$[Z + (\omega + k^2)I] \bar{A} = 0 \quad (8)$$

From the work of Ferrers and of Garbieri [8]

$$\det[Z + (\omega + k^2)I] = (\omega + k^2 - \epsilon)^2(\omega + k^2 + 2\epsilon) \quad (9)$$

Thus eq. (8) has a *symmetric* eigenvector

$$\bar{A}_s = \begin{bmatrix} A_s \\ A_s \\ A_s \end{bmatrix}$$

at $\omega + k^2 = -2\epsilon$, and two linearly independent *unsymmetric* eigenvectors at $\omega + k^2 = \epsilon$ which can be written as

$$\bar{A}_{u1} = \begin{bmatrix} A_{u1} \\ A_{u1} e^{2\pi i/3} \\ A_{u1} e^{-2\pi i/3} \end{bmatrix} \quad \text{and} \quad \bar{A}_{u2} = \begin{bmatrix} A_{u2} \\ A_{u2} e^{-2\pi i/3} \\ A_{u2} e^{2\pi i/3} \end{bmatrix}$$

A general solution of CLS is then

$$\bar{A} = \begin{bmatrix} A_s \\ A_s \\ A_s \end{bmatrix} \int_{-\infty}^{\infty} F(k) \exp[ikx - (k^2 + 2\epsilon)t] dk$$

* From here on a subscript notation is used to indicate partial differentiation.

$$\begin{aligned}
& + \begin{bmatrix} A_{u1} \\ A_{u1} e^{2\pi i/3} \\ A_{u1} e^{-2\pi i/3} \end{bmatrix} \int_{-\infty}^{\infty} G(k) \exp i[kx - (k^2 - \epsilon)t] dk \\
& + \begin{bmatrix} A_{u2} \\ A_{u2} e^{-2\pi i/3} \\ A_{u2} e^{2\pi i/3} \end{bmatrix} \int_{-\infty}^{\infty} H(k) \exp i[kx - (k^2 - \epsilon)t] dk \quad (10)
\end{aligned}$$

In the long wavelength limit ($k \rightarrow 0$) interaction of the symmetric mode with either of the unsymmetric modes will generate sinusoidal terms of the form $\exp(3\epsilon it)$. In general, however, initial conditions will require the solution to be distributed over a band of k values which in turn implies non-periodic behavior in time.

4. Coupled mode solitons on CNLS and CNLSNB

In this section we seek solitary wave (or "soliton") solutions of coupled nonlinear Schrödinger equations. Beginning with CNLS and introducing the transformation

$$\bar{A} = \bar{\phi} e^{i\omega t} \quad (11)$$

where $\bar{\phi} = \text{col}(\phi_1, \phi_2, \phi_3)$ one finds the condition

$$\det[Z + \omega I] = 0 \quad (12)$$

Equation (12) has a single root at $\omega = -2\epsilon$ corresponding to a symmetric solitary wave

$$\bar{A}_s = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \phi e^{-2i\epsilon t} \quad (13)$$

where

$$i\phi_t + \phi_{xx} + |\phi|^2 \phi = 0 \quad (14)$$

Equation (12) also has a double root at $\omega = \epsilon$ which correspond to two unsymmetric solitary waves

$$\bar{A}_{u1} = \begin{bmatrix} 1 \\ e^{2\pi i/3} \\ e^{-2\pi i/3} \end{bmatrix} \phi e^{i\epsilon t} \quad \text{and} \quad \bar{A}_{u2} = \begin{bmatrix} 1 \\ e^{-2\pi i/3} \\ e^{2\pi i/3} \end{bmatrix} \phi e^{i\epsilon t} \quad (15a, b)$$

where, again, ϕ is a soliton solution of eq. (14).

It is of particular interest to consider mixed solitary waves with symmetric and unsymmetric components for these display pulsations in the magnitudes of the spine amplitudes. For example taking

$$\begin{aligned}
\bar{A} &= a\bar{A}_{u1} + \bar{A}_s \\
&= \begin{bmatrix} a e^{i\epsilon t} + e^{-2i\epsilon t} \\ a e^{i\epsilon t + 2\pi i/3} + e^{-2i\epsilon t} \\ a e^{i\epsilon t - 2\pi i/3} + e^{-2i\epsilon t} \end{bmatrix} \phi_m
\end{aligned} \quad (16)$$

leads to the requirements that ϕ_m must satisfy the three pde's:

$$\begin{aligned}
i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2 \phi_m [1 + a^2 + 2a \cos 3\epsilon t] &= 0 \\
i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2 \phi_m [1 + a^2 + 2a \cos(3\epsilon t + 2\pi/3)] &= 0 \\
i\phi_{m,t} + \phi_{m,xx} + |\phi_m|^2 \phi_m [1 + a^2 + 2a \cos(3\epsilon t - 2\pi/3)] &= 0
\end{aligned} \quad (17a, b, c)$$

Thus this mixed mode satisfies CNLS only approximately for $|a| \ll 1$ or $|a| \gg 1$.

If, however, the mixed mode (16) is substituted into CNLSNB, eqs. (17) become

$$i\phi_{m,t} + \phi_{m,xx} + (1 + a^2)|\phi_m|^2 \phi_m = 0 \quad (18)$$

so

$$\phi_m = \frac{A_0}{\sqrt{1 + a^2}} \text{sech} \left(\frac{A_0(x - st)}{\sqrt{2}} \right) \exp i \left(\frac{s}{2} x + \frac{A_0^2}{2} t \right) \quad (19)$$

A somewhat more compact analysis of CNLSNB proceeds by introducing the transformation

$$\bar{A} = M(t) \bar{\phi} \quad (20)$$

with additional constraints

$$M^T M^* = 3I \quad (21)$$

and

$$iM_t = ZM \quad (22)$$

Under eq. (20) CNLSNB reduces to

$$i\bar{\phi}_t + \bar{\phi}_{xx} + (\bar{\phi}^* \cdot \bar{\phi}) \bar{\phi} = 0 \quad (23)$$

Assuming the columns of M to have time dependence as $\exp(i\omega t)$, eq. (22) implies

$$[Z + \omega I]M = 0 \quad (24)$$

which, with eq. (12), leads to

$$M = \begin{bmatrix} 1 & 1 & e^{-3i\epsilon t} \\ e^{2\pi i/3} & e^{-2\pi i/3} & e^{-3i\epsilon t} \\ e^{-2\pi i/3} & e^{2\pi i/3} & e^{-3i\epsilon t} \end{bmatrix} e^{i\epsilon t} \quad (25)$$

Manakov [9] has shown that eq. (23) is completely integrable. Thus the "mixed solitary wave" given by eqs. (16) and (19) [or from eq. (23) by taking $\bar{\phi} = \text{col}(a\phi_m, 0, \phi_m)$] is a true soliton of CNLSNB. The squared amplitudes (i.e., bond energies) of its three components are:

$$\begin{aligned}
|A_1|^2 &= \left[\frac{A_0^2}{1 + a^2} \text{sech}^2 \frac{A_0(x - st)}{\sqrt{2}} \right] [1 + a^2 + 2a \cos 3\epsilon t] \\
|A_2|^2 &= \left[\frac{A_0^2}{1 + a^2} \text{sech}^2 \frac{A_0(x - st)}{\sqrt{2}} \right] \\
&\quad \times [1 + a^2 + 2a \cos(3\epsilon t + 2\pi/3)] \\
|A_3|^2 &= \left[\frac{A_0^2}{1 + a^2} \text{sech}^2 \frac{A_0(x - st)}{\sqrt{2}} \right] \\
&\quad \times [1 + a^2 + 2a \cos(3\epsilon t - 2\pi/3)]
\end{aligned} \quad (26a, b, c)$$

Following Gibbons et al. [10] it is straightforward to show that the Hamiltonian associated with CNLSNB is

$$\begin{aligned}
H &= \int_{-\infty}^{\infty} \left\{ \sum_{j=1}^3 |A_{j,x}|^2 - \frac{1}{6} \left[\sum_j |A_j|^2 \right]^2 + \epsilon(A_1^* A_2 + A_1 A_2^* \right. \\
&\quad \left. + A_1^* A_3 + A_1 A_3^* + A_2^* A_3 + A_2 A_3^*) \right\} dx
\end{aligned} \quad (27)$$

while the "mass"

$$m = \frac{1}{2} \int_{-\infty}^{\infty} [|A_1|^2 + |A_2|^2 + |A_3|^2] dx \quad (28)$$

Thus the energy of the soliton is

$$H = \frac{1}{2}ms^2 - \frac{m^3}{54} + 2m\epsilon \left(\frac{2-a^2}{1+a^2} \right) \quad (29)$$

Physical significance is readily assigned to the terms in eq. (29): the first term is kinetic energy, the second is binding energy, and the third is interspine interaction energy. It is energetically favorable for the mixed mode to add more of the unsymmetric mode by increasing a .

5. Numerical results on the modified Davydov equations

In this section we turn to numerical results obtained from integration of the MD equations. As in [5] the boundary conditions are

$$A_{n\alpha} \text{ and } B_{n\alpha} = 0 \text{ for: } n = 0, -1, -2, -3,$$

$$n = n_{\max+1}, n_{\max+2}, n_{\max+3}, \text{ and all } \tau$$

The initial condition (at $\tau = 0$) is

$$A_{11} = 1 \quad \text{and} \quad A_{12} = 1$$

$$\text{other } A_{n\alpha} = 0$$

$$\text{all } B_{n\alpha} = 0$$

With $n_{\max} = 200$ and $\chi_1 = 0.3 \times 10^{-10}$ newtons (i.e., just below threshold), Figure 3 shows the peak amplitudes on each of the three spines as a function of τ . Figure 4 shows a corresponding results for $\chi_1 = 0.4 \times 10^{-10}$ newtons (just above threshold). A well developed interspine oscillation is seen in Fig. 4 with a period (in computer units) of

$$T_1 = 21.93 \pm 0.806 \quad (30)$$

where the "±" refers to r.m.s. deviations in a series of measurements. The time dependence in Fig. 4 is qualitatively similar to that derived in eq. (26), but from eq. (5) and Table I, $\epsilon = 0.033$ which implies a period $(3\epsilon)^{-1} = 10$. This disagreement is not unexpected considering the approximations made in going from MD through Z and CNLS to CNLSNB.

The speed of the solitary wave under the conditions of Fig. 4 is

$$s = 0.3733 \pm 0.0189. \quad (31)$$

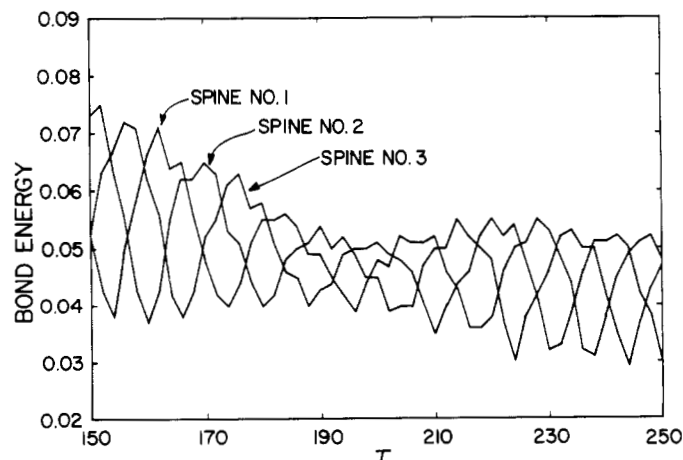


Fig. 3. Peak bond energy on each spine as a function of computer time with the nonlinear parameter $\chi_1 = 0.3 \times 10^{-10}$ newtons which is below threshold for soliton formation.

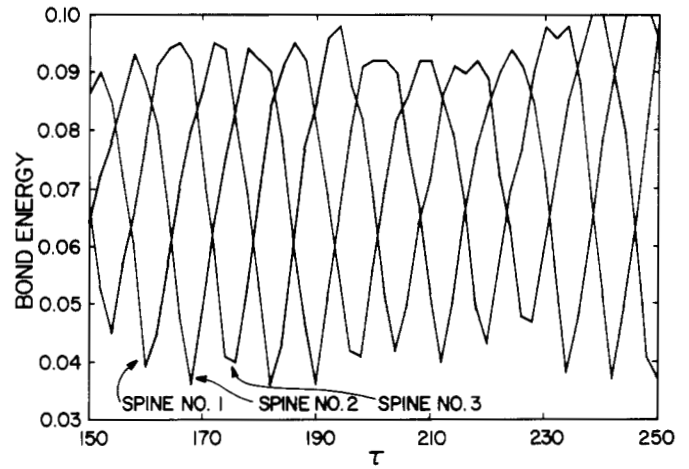


Fig. 4. Same as Fig. 3 but $\chi_1 = 0.4 \times 10^{-10}$ newtons which is above threshold for soliton formation.

Assuming the bond energies on each spine to be described by eq. (26), the mixing parameter, a , is

$$a = 0.22 \quad \text{at } \tau = 170$$

$$a = 0.26 \quad \text{at } \tau = 412$$

Thus there is a rather small tendency for the unsymmetric mode component to increase with τ .

As Fig. 3 shows, the situation below threshold is quite different. Although there is some tendency toward an oscillation of period T_1 , the bond amplitude variations are much smaller and the frequency is less uniform. Further appreciation of the dynamic differences below and above threshold can be had by comparing the spatial shapes of the waves in Figs. 5 and 6. Not only is the oscillation frequency more variable below threshold, but a smaller fraction of the total bond energy participates in it.

6. Interpretation of laser-Raman spectral measurements as internal vibrations of alpha-helix solitons

The interspine oscillation shown in Fig. 4 has a spectral energy

$$E_1 = \frac{\sqrt{\omega/M}}{cT_1} \text{ cm}^{-1} \quad (32)$$

where numerical values for $\sqrt{\omega/M}$ and T_1 are given in eqs. (3) and (30) respectively, and c is the speed of light. Thus

$$E_1 = 15.35(1 \pm 0.0368) \text{ cm}^{-1} \quad (33)$$

A second oscillation is generated with a period equal to the time the soliton takes to pass by a unit cell. This is s^{-1} in com-

Table II

Measured (cm^{-1})	Assignments
45	$2E_1$
52	$3E_1$
63	$3E_1$
85	$5E_1$
90	$E_2 - 2E_1$
108	$E_2 - E_1$
123	E_2
152	$E_2 + 2E_1$
182	$E_2 + 3E_1$

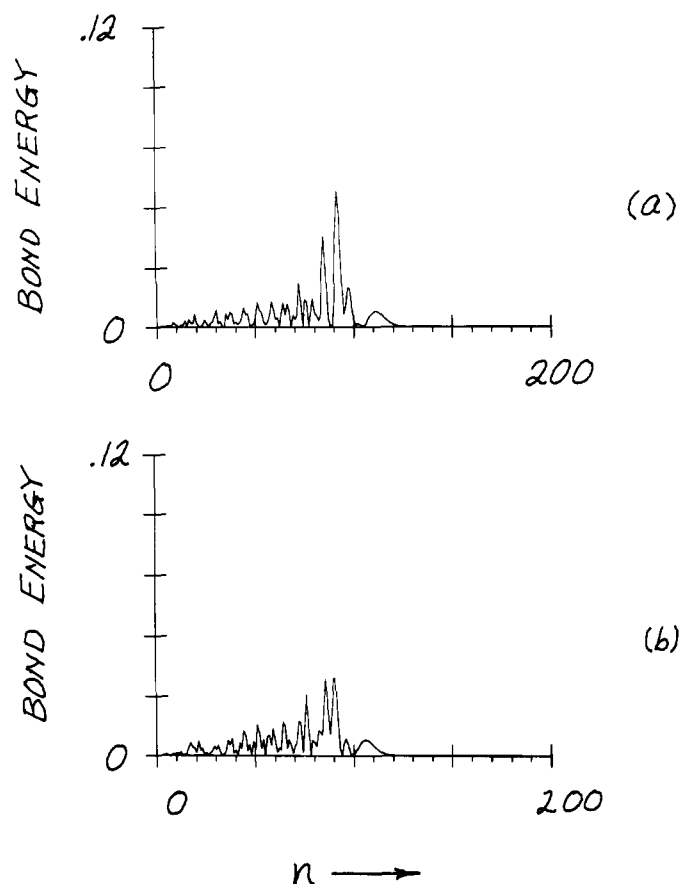


Fig. 5. Spatial wave form on spin No. 2 below threshold ($x_1 = 0.3 \times 10^{-10}$ newtons). (a) Maximum amplitude at $\tau = 234$. (b) Minimum amplitude at $\tau = 224$.

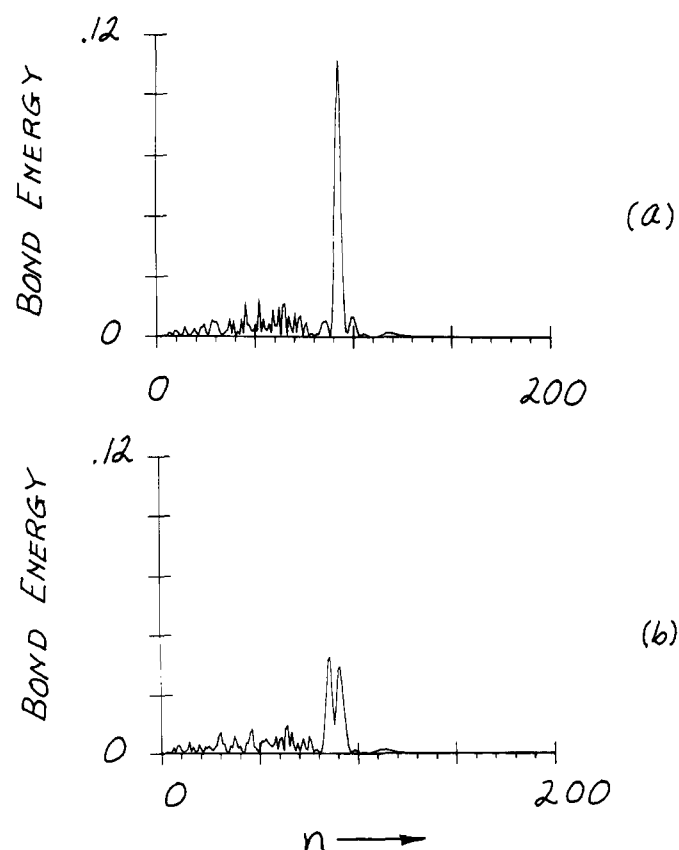


Fig. 6. Spatial waveform on spine No. 2 above threshold ($x_1 = 0.4 \times 10^{-10}$ newtons). (a) Maximum amplitude at $\tau = 246$. (b) Minimum amplitude at $\tau = 234$.

puter units, where s is given in eq. (31), so the corresponding spectral energy is

$$E_2 = 125.7(1 \pm 0.0506) \text{ cm}^{-1} \quad (34)$$

In [7] the spectral lines obtained by Webb from metabolically active *E. coli* [6] (see Fig. 2) were tentatively assigned as simple sums and differences of E_1 and E_2 as shown in Table II. With these assignments and the numerical values for E_1 and E_2 given in eqs. (33) and (34), the relation between calculated and measured line energies is plotted in Fig. 7. Broadly speaking, the differences between measurements and calculations can be attributed to errors in the numerical model (MD) and the measurement errors. With the line assignments as in Table II, the best that the numerical model can do is a least squares fit which (on integer values) is at

$$E_1 = 17 \text{ cm}^{-1} \quad \text{and} \quad E_2 = 124 \text{ cm}^{-1} \quad (35)$$

Assuming the three parameters in MD to be such that the fundamental energies are given by eq. (35) rather than eqs. (33) and (34) (a 10% and a 1.3% change respectively), the relation between measured and calculated lines is plotted again in Fig. 8. There are three possible explanations for the differences between measurements and calculations in Fig. 8: (i) The theory presented here is wrong, (ii) The line assignments are incorrect, and (iii) Measurement errors. Only the first line (measured at 45 cm^{-1}) lies significantly more than a spectral bandwidth from agreement. From Fig. 2 it is seen that this line rests high on the background shoulder (of Rayleigh scattering) which may introduce additional errors into the measurement.

Further support for the hypothesis presented here is provided by the data in Fig. 9. This is a histogram of 125 lines obtained from ten experiments done by S. J. Webb and M. E. Stoneham between 1975 and 1979 [11]. With the assignments as indicated, a least squares fit to these data requires

$$E_1 = 16.8 \text{ cm}^{-1} \quad \text{and} \quad E_2 = 126.0 \text{ cm}^{-1} \quad (36)$$

Spectral lines calculated from these frequencies are also indicated on Fig. 9.

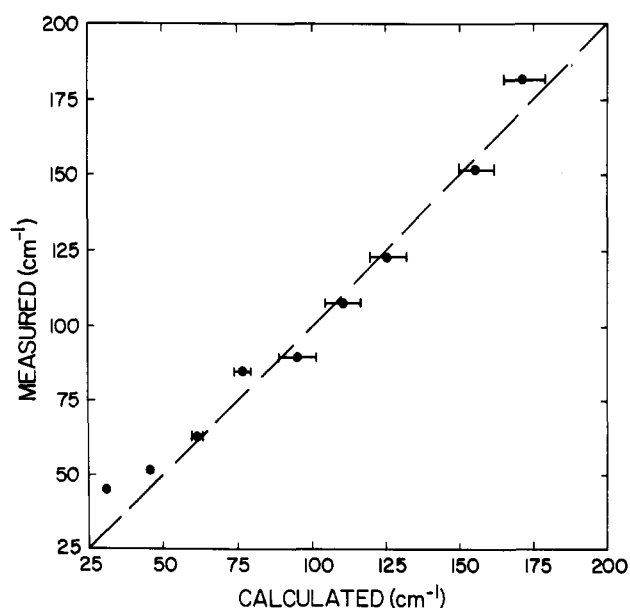


Fig. 7. Comparison laser-Raman lines measured in Fig. 2 with calculations according to the assignments in Table II from E_1 in eq. (33) and E_2 in eq. (34).

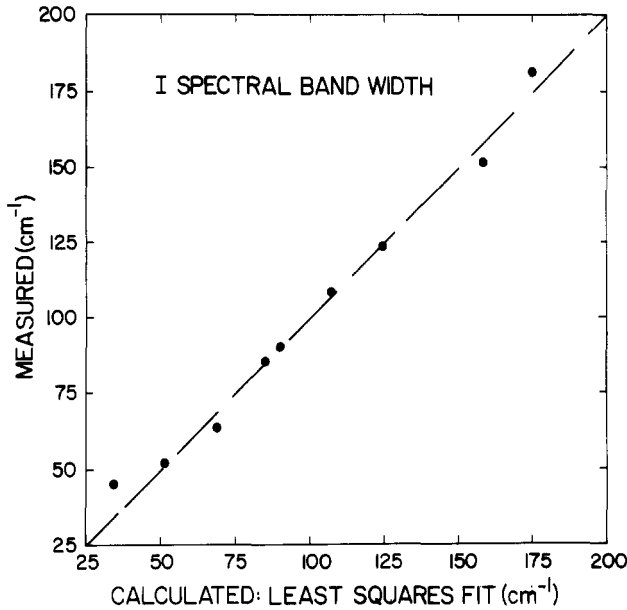


Fig. 8. Same as Fig. 7 but E_1 and E_2 are the "least-squares-fit" given in eq. (35).

7. Concluding discussion

The central question in this paper is to decide whether the laser-Raman spectrum displayed in Fig. 2 can be assigned to internal oscillations of alpha-helix solitons. According to Webb [6] these spectra are characterized by the following facts: (i) At 300 K a Raman spectrum is observed only when the cells are metabolically active, (ii) The intensity ratios of Stokes to anti-Stokes lines indicate that the Raman active states are produced *in vivo* by non-thermal means, and (iii) Spectral lines below 300 cm^{-1} shift to lower wave numbers as the cells progress through their life cycles. Although broad laser-Raman spectra have been observed on the alpha-helix proteins in aqueous solution and in crystalline form [12–14], such an explanation is unlikely for the lines in Fig. 2.

A convenient summary of the foregoing results is provided by Table III.

Since there was no adjustment of parameters in MD prior to the calculations of row (1), the agreement displayed in this table is quite remarkable. There is little difficulty in supposing the MD

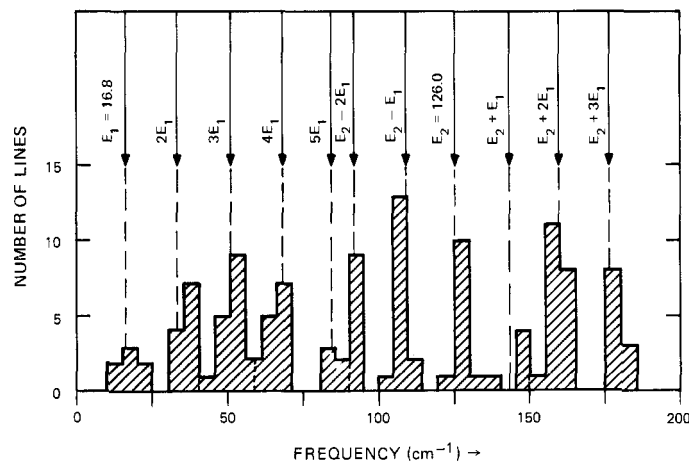


Fig. 9. Histogram of lines from ten laser-Raman experiments on metabolically active *E. coli*. Calculated lines are from the least squares fit values of E_1 and E_2 given in eq. (36). Data from Webb, S. J. (private communication).

Table III

Item	$E_1(\text{cm}^{-1})$	$E_2(\text{cm}^{-1})$
(1) Calculation of component frequencies from MD.	$15.4 \pm 3.7\%$	$125.7 \pm 5.1\%$
(2) Least squares fit (over integer values) to lines in Fig. 2.	17	124
(3) Least squares fit to data in Fig. 9.	16.8	126.0

parameters to be slightly changed to obtain the fundamental energies listed in row (2) or row (3). Since E_2 is related to the frequency at which a soliton interferes with the alpha-helix periodicity, it could be expected to change with the soliton speed under varying metabolic conditions.

Figure 8 shows the agreement between the lines measured from the spectrum of Fig. 2 and those calculated from the energies listed in the second row of Table III. Figure 9 shows the agreement between data from 125 lines measured during ten separate experiments and those calculated from the energies listed in the third row of Table III. The differences between measured and calculated lines shown in Figs. 8 and 9 must be attributed either to measurement errors or theoretical misconceptions [15]. Noting that [11, 16]: (i) The spectral band width of the measurements is 5 cm^{-1} ; (ii) The lines often move several cm^{-1} during the time of measurement (~ 100 minutes); and (iii) Low frequency lines are difficult to resolve because of Rayleigh scattering (see Fig. 2), I conclude that Figs. 8 and 9 show reasonable agreement between measurements and calculations.

The reader may not agree, but certainly the evidence is strong enough to encourage a vigorous search for corroborative experiments.

Appendix

Here we consider a generalization of CNLSNB from three to n interacting channels. Thus

$$i\bar{A}_t + \bar{A}_{xx} + \frac{1}{n}(\bar{A}^* \cdot \bar{A})\bar{A} = Z\bar{A} \quad (\text{A-1})$$

where $\bar{A} = \text{col}(A_1, \dots, A_n)$. If the channels are spaced close together compared with the decay constant for lateral interaction, it is reasonable to assume

$$Z = \begin{bmatrix} 0 & \epsilon & \epsilon & \dots & \epsilon \\ \epsilon & 0 & \epsilon & \dots & \epsilon \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \epsilon & \cdot & \cdot & \epsilon & 0 \end{bmatrix}_{n \times n} \quad (\text{A-2})$$

The transformation (20) with constraints

$$M^T M^* = nI \quad (\text{A-3})$$

and

$$iM_t = ZM \quad (\text{A-4})$$

reduces (A-1) to

$$i\bar{\phi}_t + \bar{\phi}_{xx} + (\bar{\phi}^* \cdot \bar{\phi})\bar{\phi} = 0 \quad (\text{A-5})$$

where $\bar{\phi} = \text{col}(\phi_1, \dots, \phi_n)$.

From (A-4) the time dependence of columns of M is as $\exp(i\omega t)$ where ω satisfies $\det[Z + \omega I] = 0$. The general Ferrers–Garbieri formula for this determinant gives [8]

$$(\omega - \epsilon)^{n-1}[\omega + (n-1)\epsilon] = 0 \quad (\text{A-6})$$

Thus one column of M has time dependence $\exp[i(n-1)\epsilon t]$ and has equal coefficients. The remaining columns have time dependence $\exp(i\epsilon t)$ and are on an orthogonal set in the null space of $Z + \epsilon I$.

Manakov [9] has given Lax operators [17] for eq. (A-5). Thus it is completely integrable using the inverse scattering transform method [18] and it has soliton solutions which, upon transformation by eq. (20), can be interpreted as solutions of eq. (A-1).

If the n channels are not spaced closely with respect to the decay constant for lateral interaction, the non-diagonal elements in Z will not be equal and the Ferrers–Garbieri formula cannot be used to factor $\det[Z + \omega I]$. For four channels arranged at the corners of a square, however,

$$Z = \begin{bmatrix} 0 & \epsilon_1 & \epsilon_1 & \epsilon_2 \\ \epsilon_1 & 0 & \epsilon_2 & \epsilon_1 \\ \epsilon_1 & \epsilon_2 & 0 & \epsilon_1 \\ \epsilon_2 & \epsilon_1 & \epsilon_1 & 0 \end{bmatrix} \quad (\text{A-11})$$

and

$$\det[Z + \omega I] = (\omega - \epsilon_2)^2(\omega + \epsilon_2 - 2\epsilon_1)(\omega + \epsilon_2 + 2\epsilon_1) \quad (\text{A-12})$$

Acknowledgements

It is a pleasure to thank Professor S. J. Webb for correspondence about the accuracy of laser-Raman measurements and for providing the data that is summarized in Fig. 9.

References

1. Davydov, A. S. and Kislukha, N. I., *Phys. Stat. Sol. (b)* **59**, 465 (1973). See Davydov, A. S., *Physica Scripta* **20**, 387 (1979); and *Biology and Quantum Mechanics*, Pergamon Press (1982) for extensive bibliographies.
2. Pauling, L., *The Nature of the Chemical Bond*. Cornell U. Press (1960).
3. Hyman, J. M., McLaughlin, D. W. and Scott, A. C., *Physica* **3D**, 23 (1981).
4. Private communication from Kuprievich, V. A. and Kudritskaja, Z. G., October 5, 1980.
5. Scott, A. C., *Phys. Rev. B* (to appear).
6. Webb, S. J., *Physics Reports* **60**, 201 (1980).
7. Scott, A. C., *Phys. Lett.* **86A**, 60 (1981).
8. Muir, T., *The Theory of Determinants*. MacMillan, London: Vol. II (1911) pp. 141–142; Vol. III (1920) pp. 109–110.
9. Manakov, S. V., *Sov. Phys. – JETP* **38**, 248 (1974). See also Kulish, P. P. and Sklyanain, E. K., *Phys. Lett.* **84A**, 349 (1981).
10. Gibbons, J., Thornhill, S. G., Wardrop, M. J. and ter Haar, D., *J. Plasma Physics* **17**, 153 (1977).
11. Private communication from Webb, S. J., November 20, 1981.
12. Small, E. W., Fanconi, B. and Peticolas, W. L., *J. Chem. Phys.* **52**, 4369 (1970).
13. Brown, K. G., Erfurth, S. C., Small, E. W. and Peticolas, W. L., *Proc. Nat. Acad. Sci. USA* **69**, 1467 (1972).
14. Genzel, L., Keilman, F., Martin, T. P., Winterling, G., Yacoby, Y., Frölich, H. and Makinen, M. W., *Biopolymers* **15**, 219 (1976).
15. For example, Fedyanin, V. K., Makhankov, V. G. and Yakushevich, L. V., *Phys. Lett.* **61A**, 256 (1977) have shown that inclusion of resonance interaction terms of the form $\gamma \sum_n (a_n^\dagger a_{n+1}^\dagger - a_n a_{n+1})$ into the original Hamiltonian (where $a_n^\dagger$ is a creation operator for bond energy quanta at unit cell n and γ is a small parameter) introduces terms of the form $\gamma \bar{A}_x^*$ into the left hand side of CNLSNB. Even for small γ this term has qualitative effects on solitary wave behavior, some of which are discussed in Fedyanin, V. K. and Makhankov, V. G., *Physica Scripta* **20**, 552 (1979).
16. Webb, S. J. and Stoneham, M. E., *Phys. Lett.* **63A**, 407 (1977). See also Drissler, F., *Collective Phenomena* **3**, 147 (1981); and Webb, S. J., *Collective Phenomena* **3**, 313 (1981).
17. Lax, P. D., *Comm. Pure Appl. Math.* **21**, 467 (1968).
18. Gardner, C. S., Greene, J. M., Kruskal, M. D. and Miura, R. M., *Phys. Rev. Lett.* **19**, 1095 (1967).